

2026 SPACE ASTRONOMY SUMMER PROGRAM

Using Machine Learning to Discover Fading AGN

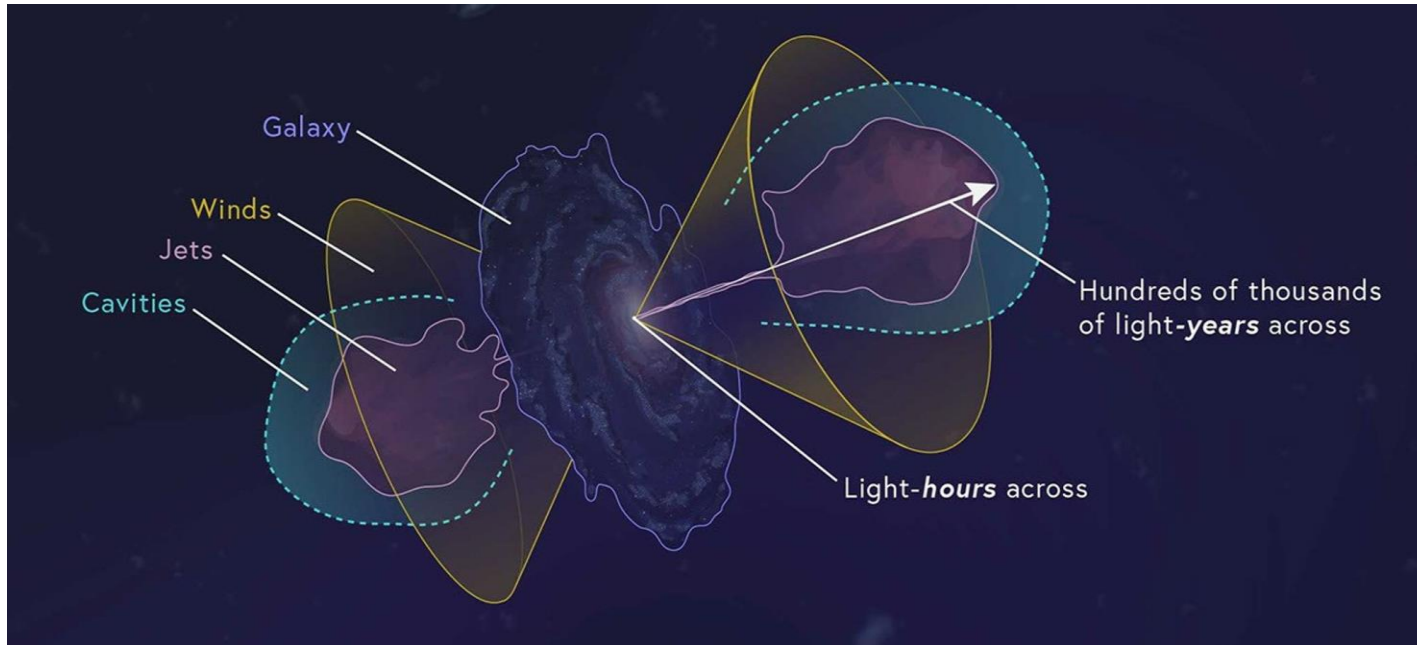
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AGN: Active Galactic Nuclei

- Active Supermassive Black Holes
- Grow by accretion
- Impact their host galaxies and their evolution



AGN Anatomy
Credit: NASA Science

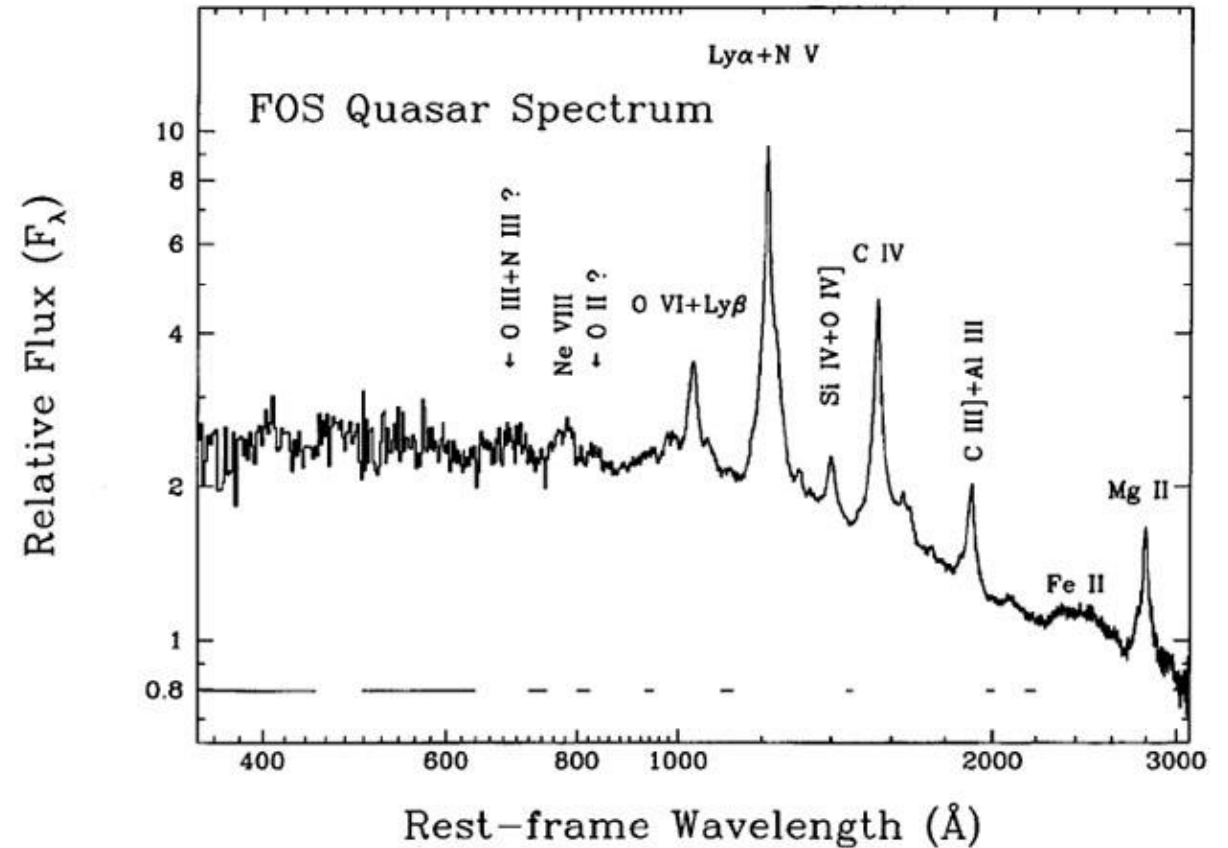


**An Artist's Illustration of
an Accretion Disk**
Credit: NASA/JPL-Caltech



Ionization lines in AGN

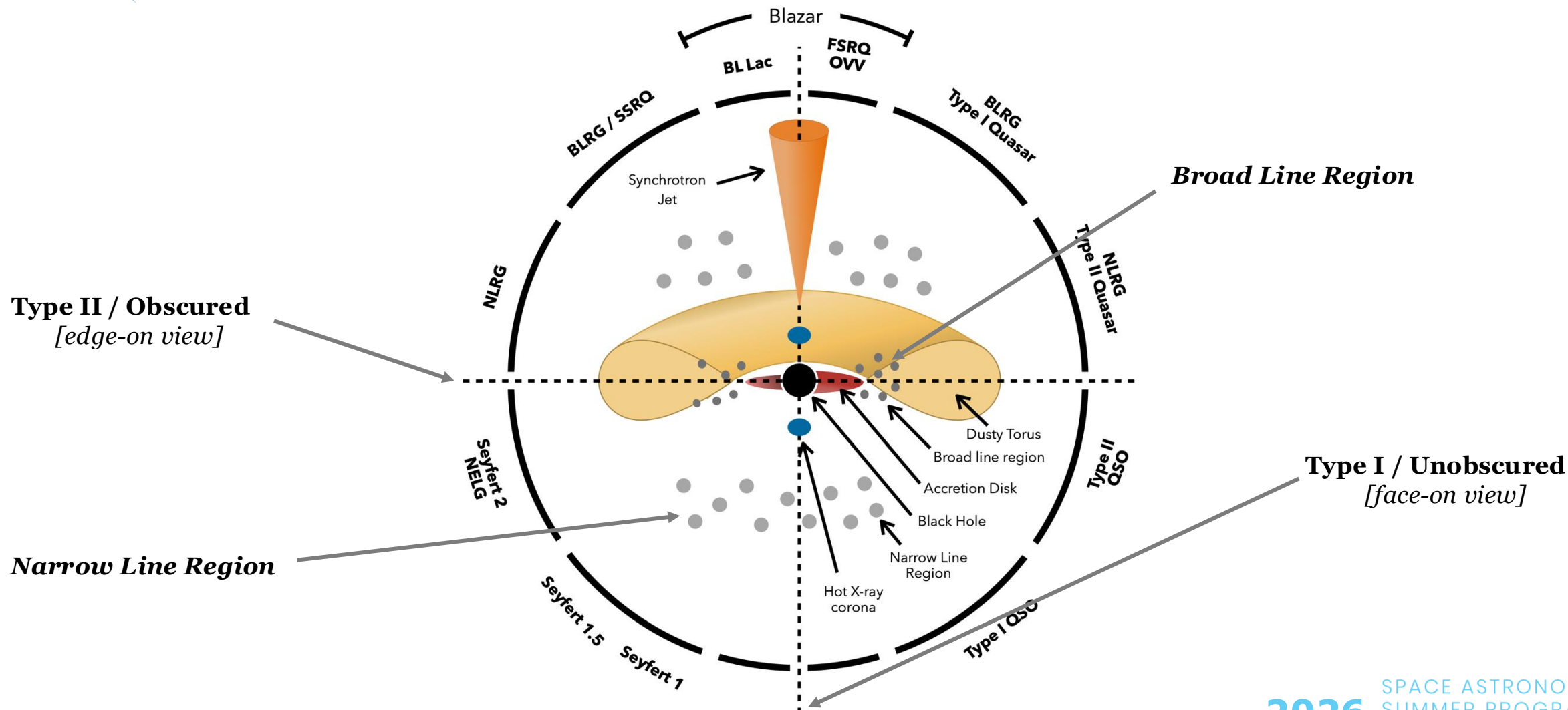
- Emission lines and absorptions lines caused by electrons transitioning between energy levels
- Because of its high energetic center, AGN have distinct emission lines from normal galaxies due to their high ionization power
- Broad lines: Balmer series, He, Fe
- Narrow lines: [O III], [N II], [S II]



Credit: Caltech.edu, Koratkar & Blaes

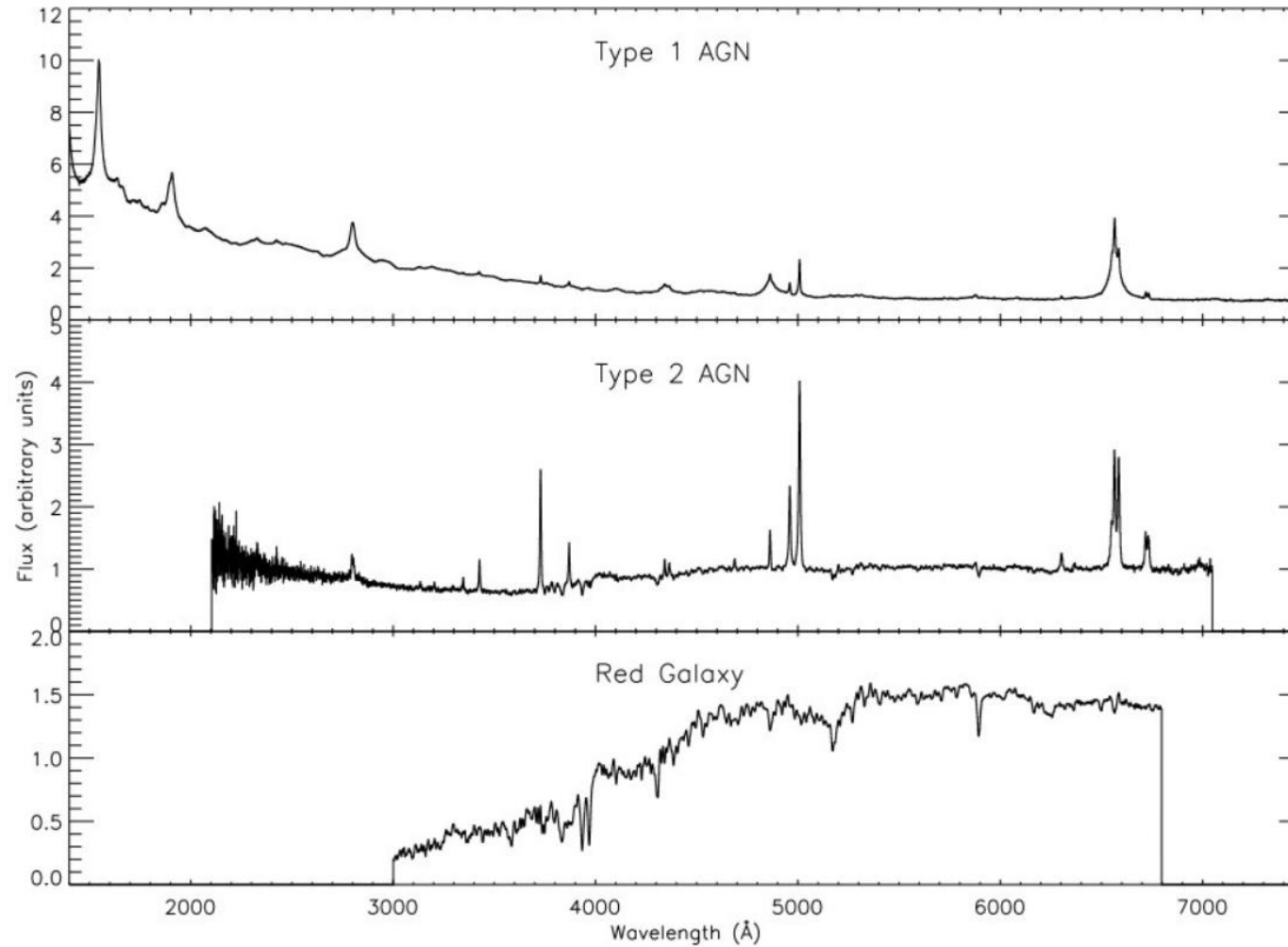


Types of AGN





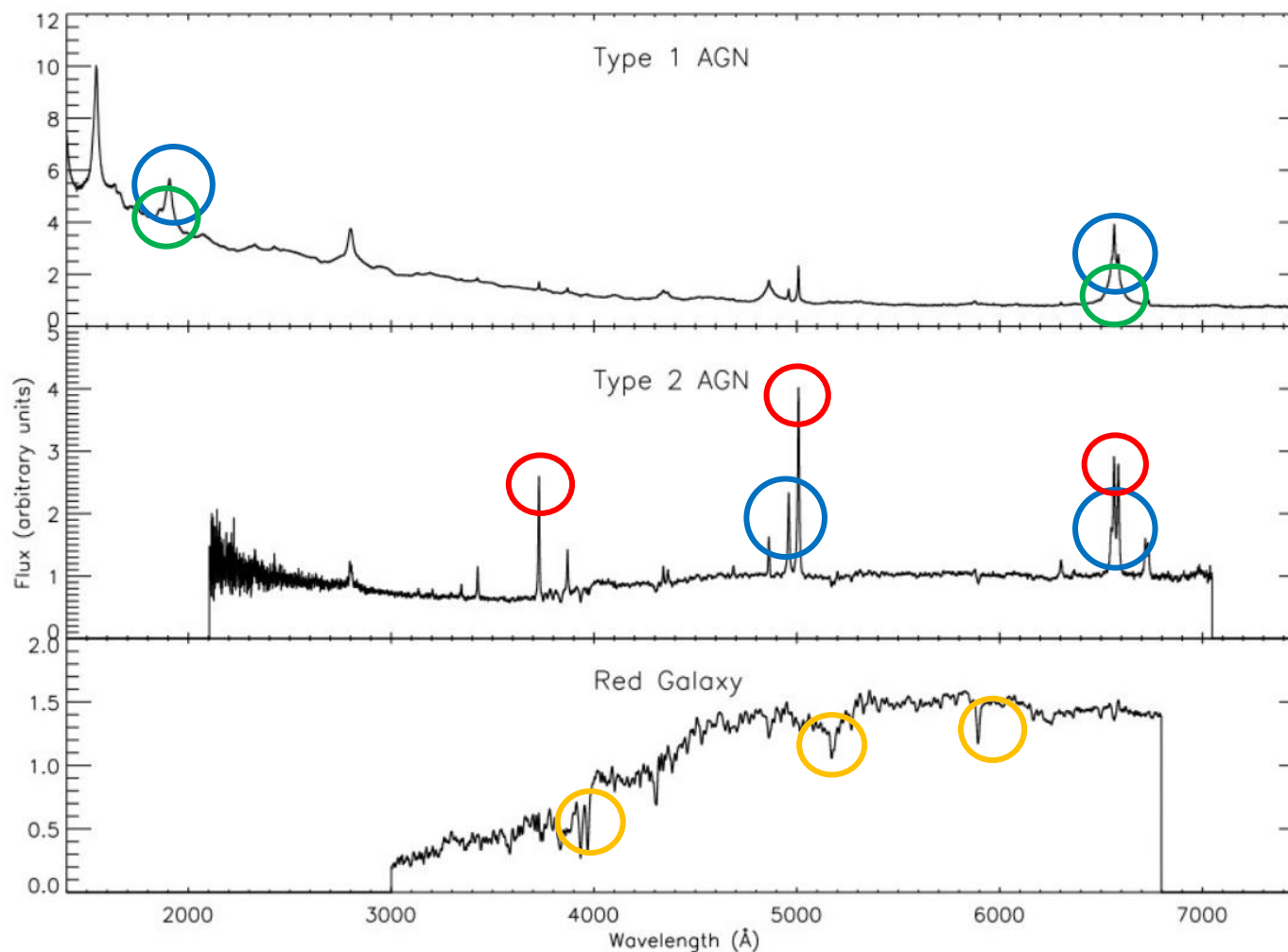
AGN Spectra



Credit: Mickaelian, 2015



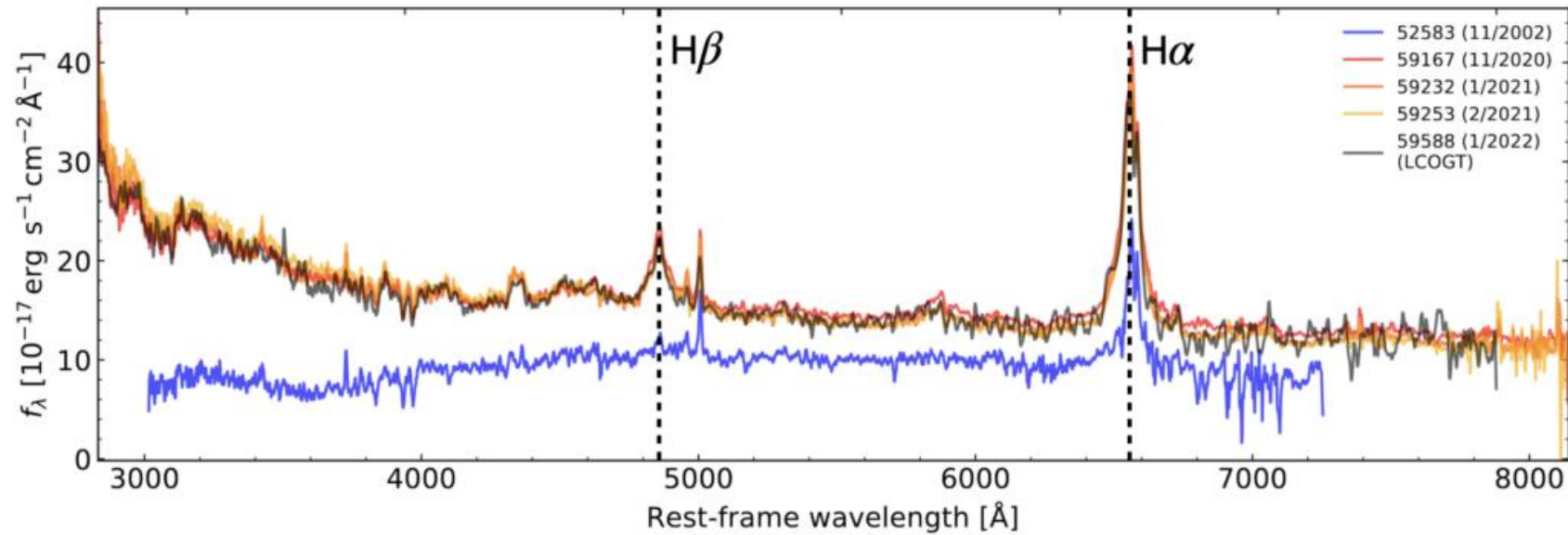
AGN Spectra



Emission Lines
Absorption Lines
Narrow Emission Lines
Broad Emission Lines



Changing Look AGN



The blue spectrum shows the original SDSS spectrum from 2002, and the other spectra are from 2021-2022, where there is now a strong broad H β line and a blue continuum.

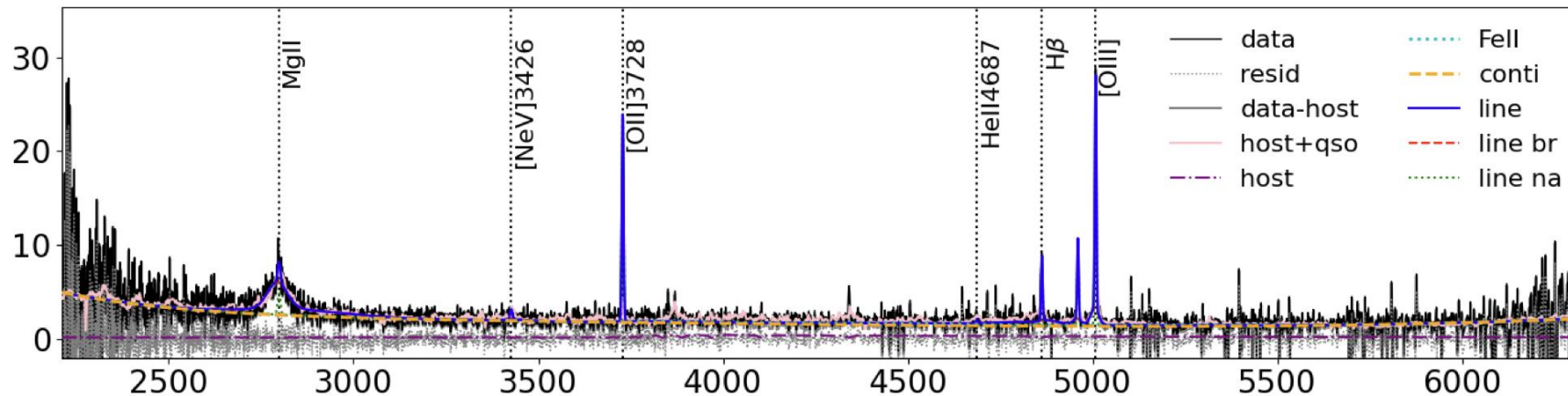
Credit: Zeltyn et. al. 2024



Fading AGN?

Key findings from LaMassa et al. (2026)

- Identified ~500 SDSS galaxies with broad MgII emission
- Unlike typical Type 1 AGN, these galaxies show:
 - weak or absent broad Balmer lines ($H\alpha$, $H\beta$)
 - despite retaining broad MgII
- Proposed that these galaxies could represent fading AGN



Spectra of a 'fading' AGN



Project Goal

- **Extremely rare:** ~500 broad MgII galaxies identified among ~1.3 million SDSS spectra
- **Goal:** Build on Steph's work to identify fading AGN candidates through broad MgII emission

Can we use machine learning to more efficiently identify broad MgII galaxies from over a million spectra?



Methodology: Machine Learning

- A machine learning classifier could efficiently identify spectral patterns associated with fading AGN
- Train a supervised machine learning model to distinguish
 - Broad MgII galaxies: ~500 sources identified in Steph's paper
 - Normal SDSS galaxies: control sample
- We are using CNN: Convolutional Neural Networks



CNN: Convolutional Neural Networks

- Spectra are one-dimensional signals, similar to time series
- **Layers** progressively learn more complex spectral features
- **Neurons** detect specific patterns in the spectrum
- The network automatically learns spectral signatures instead of requiring manually selected features
- What the CNN learns:
 - broad emission line profiles
 - flux variations around MgII
 - spectral shape differences in AGN and normal galaxies

Convolutional Layer 1
Detects simple local patterns

Max Pooling 1
Compresses the information,
retains the strongest features

Convolutional layer 2
Combines simple features into
more complex patterns

Max Pooling 2
Further compresses the data

Flattening
Feature maps → 1D vector

Fully connected layers
Combines learned features



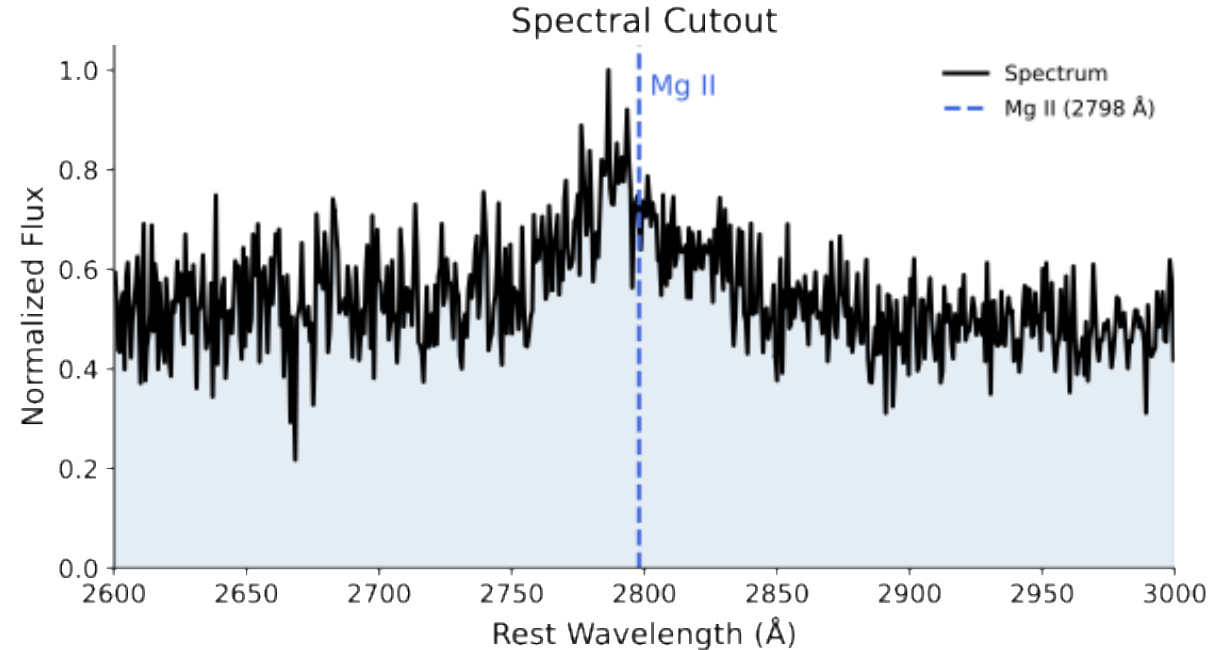
Pipeline

1. Data Preparation

- Each dataset split has same number of broad and normal spectra files:
 - Training: 80%
 - Validation: 20%
 - Testing: 20%

2. Spectral Processing

- Shift spectra to rest frame
- Extract MgII wavelength region: 2600-3000 angstroms
- Interpolate spectra onto a common wavelength grid
- Normalize flux values





Pipeline (continued)

3. CNN Training

- Train 1D CNN classifier
- Use weighted sampling to account for class balance
- Optimize classification loss

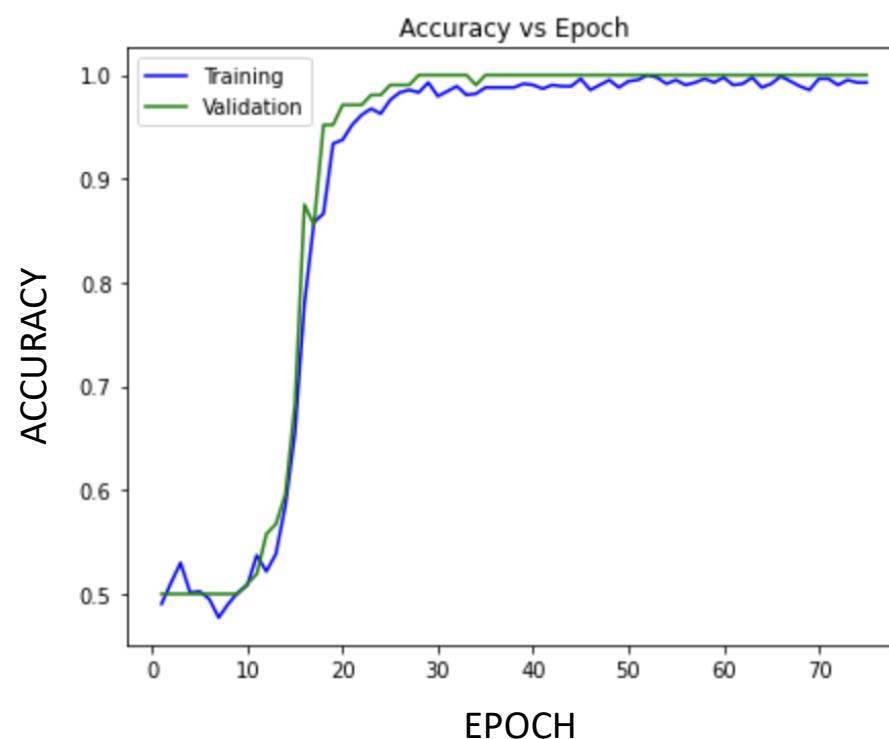
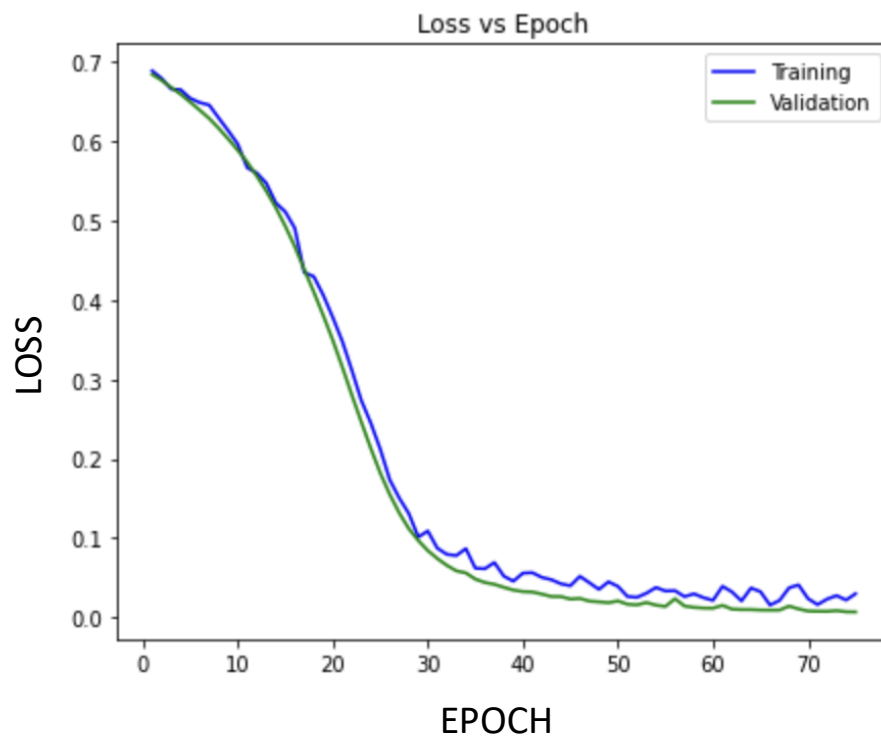
4. Model Evaluation

Evaluate performance using:

- **Accuracy:** overall correct classifications
- **Precision:** number of correct predicted positives
- **Recall:** number of actual positives found
- **Confusion Matrix:** summary of prediction results



Results



CNN Parameters: 2 layers, 32 neurons

Loss vs. Epoch: Shows how the model's prediction error changes during training (lower is better)

Accuracy vs. Epoch: Shows how the model's classification performance improves during training (higher is better)



Results

CNN Confusion Matrix
Test Dataset

True Class	Normal	Broad Mg II
Normal	52	0
Broad Mg II	1	51
Predicted Class		

Test Results

Loss=0.0336

Accuracy=0.990

Precision=1.000

Recall=0.981

True Negatives: 52

True Positives: 51

False Negative: 1

False Positives: 0



Final Work

Deployment on SDSS Dataset

- Apply trained CNN to the full SDSS galaxy sample
- Identify new broad MgII candidates
- Compare recovered candidates with known training/validation galaxies

Results

- Broad MgII Candidates: 7382
- Normal Galaxies: 1,340,345
- Total: 1,347,727



Future Work

Inspect candidate spectra to validate newly identified broad MgII galaxies

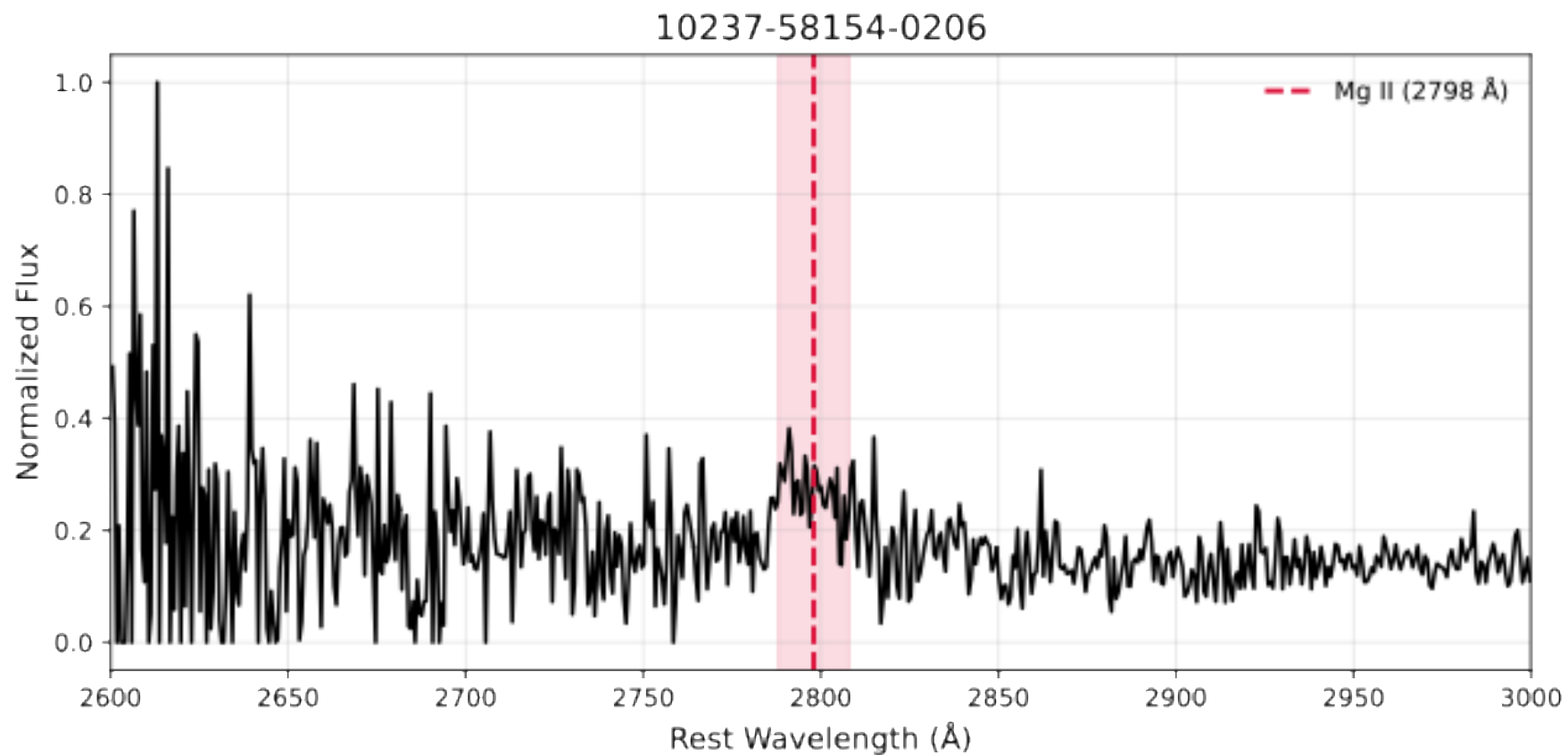
Analyze model failure modes to understand misclassifications

Characterize new candidates through detailed spectral analysis to evaluate their potential as “fading AGN”



Thank you!

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Testing dataset *False Negative Case*